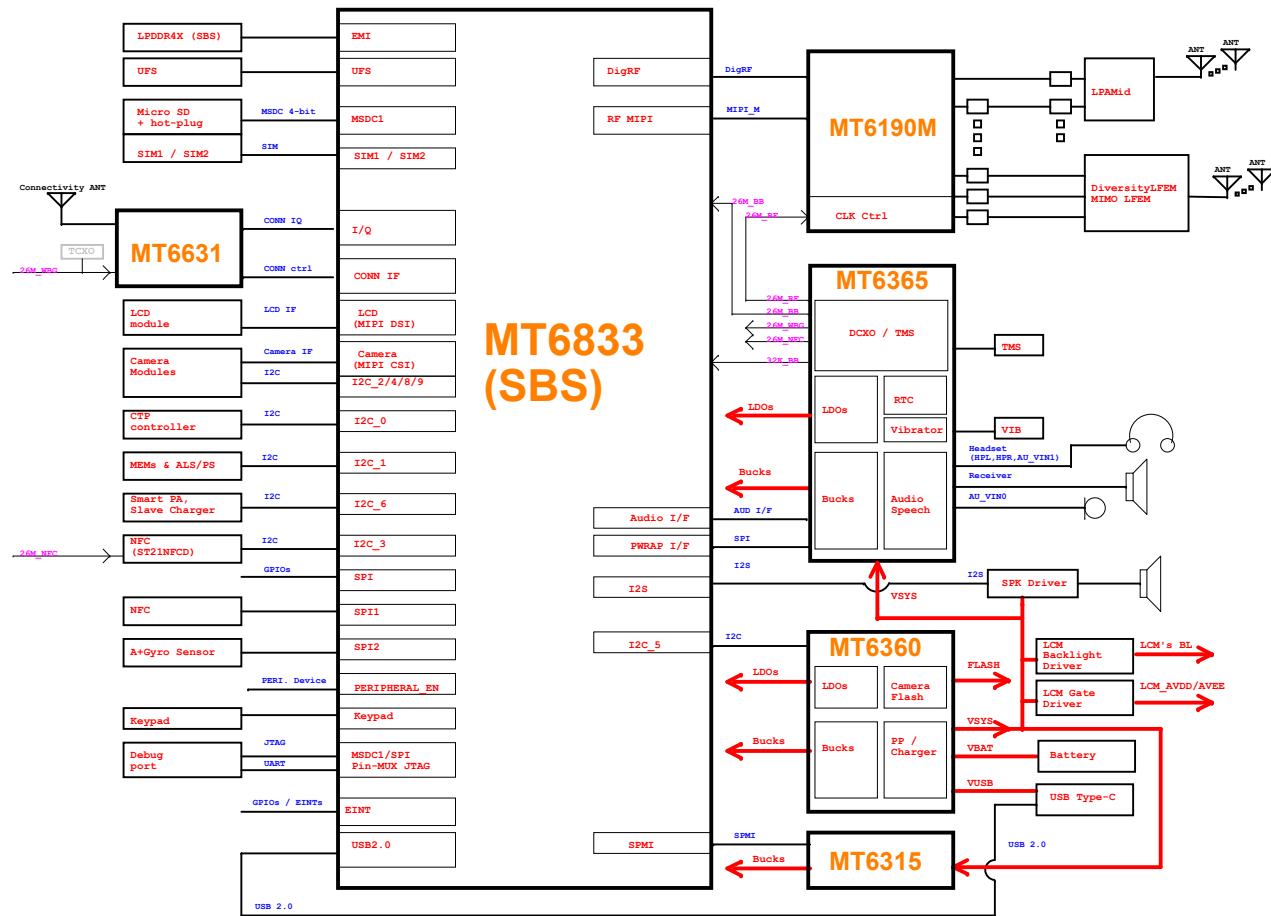
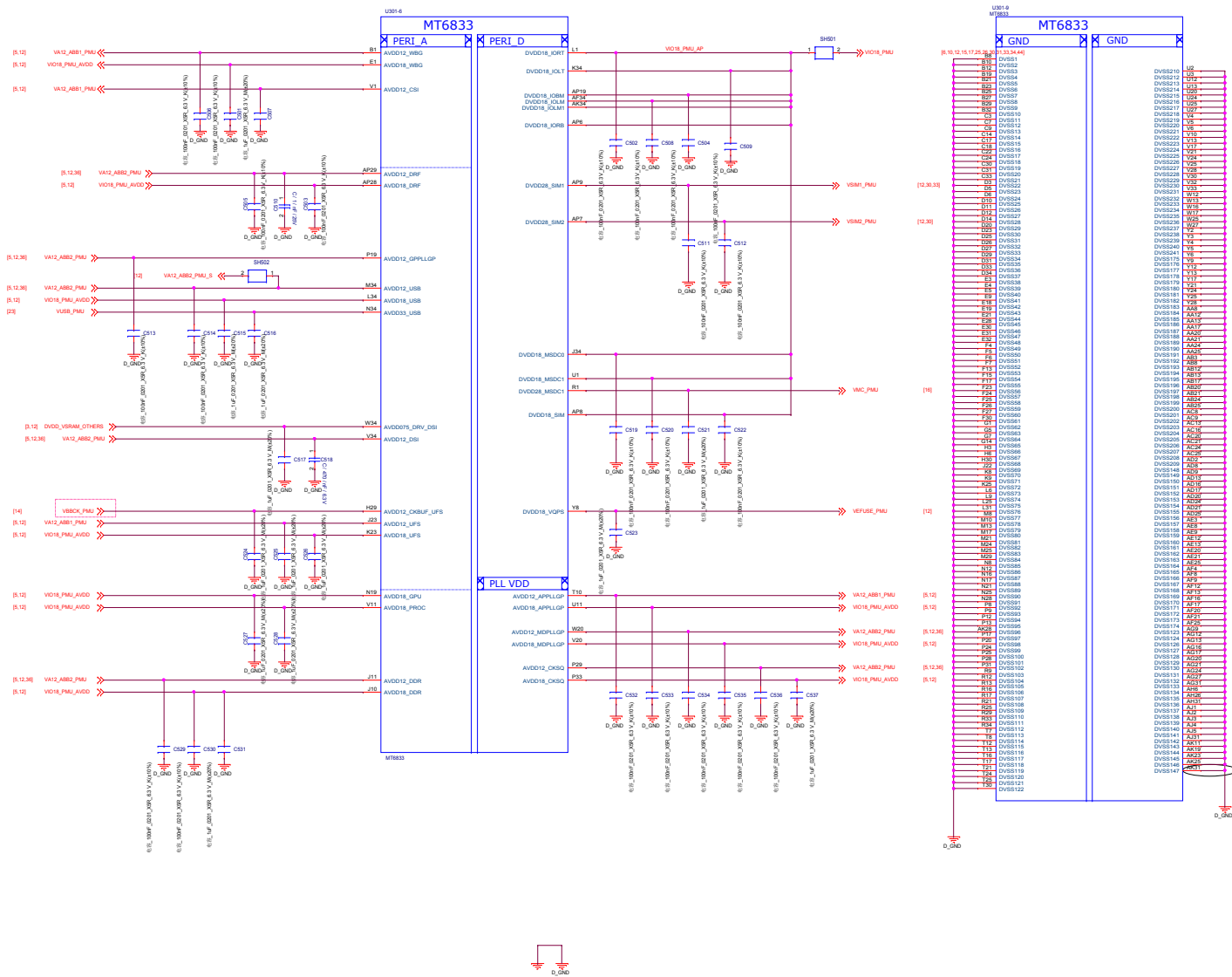




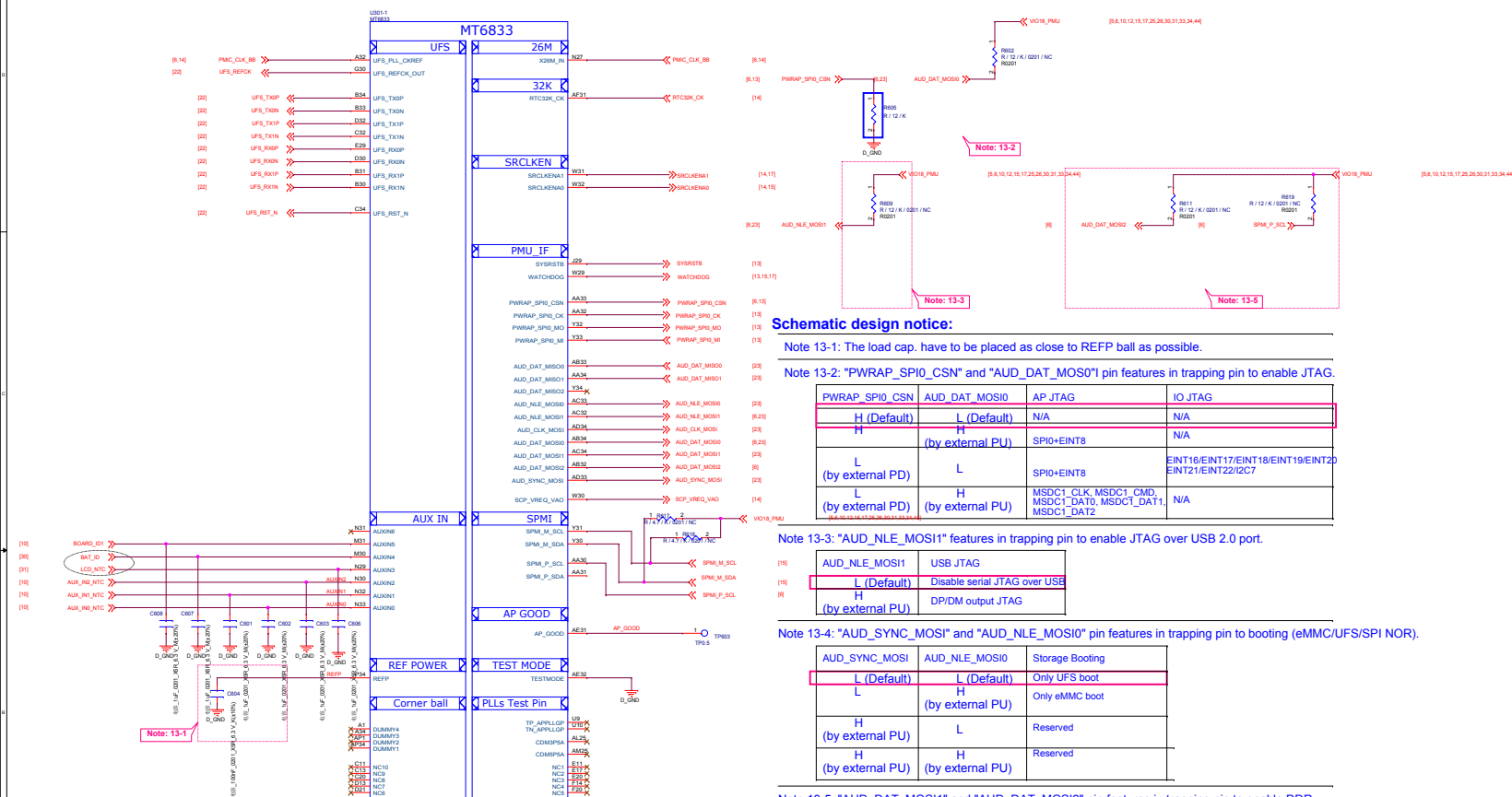
华勤通讯 Huaqin Telecom Technology Com., Ltd		
Title Block Diagram		
Size D	Project <Doc>	Rev V1
Date: Tuesday, January 12, 2021	Sheet 1 of 48	

Date	Version	Page	Change notice
2020.06.xx	Pre-Released		
2020.07.xx	V01		
2020.08.14	V02	13_BB_1 29_POWER_EXT 20_POWER_MT6365_Buck 25_POWER_MT6315_VMD	Disconnect Dummy pins A1,A34,AP1,AP34 to GND Change C2923 cap from 6.3V to 10V [PDN] Change Caps of DVDD_GPU - C2039/C2038 to NC [PDN] Change Caps of DVDD_MODEM - C2502/C2534/C2523/C2522 to NC
2020.08.26	V03	File name	Modify the file name

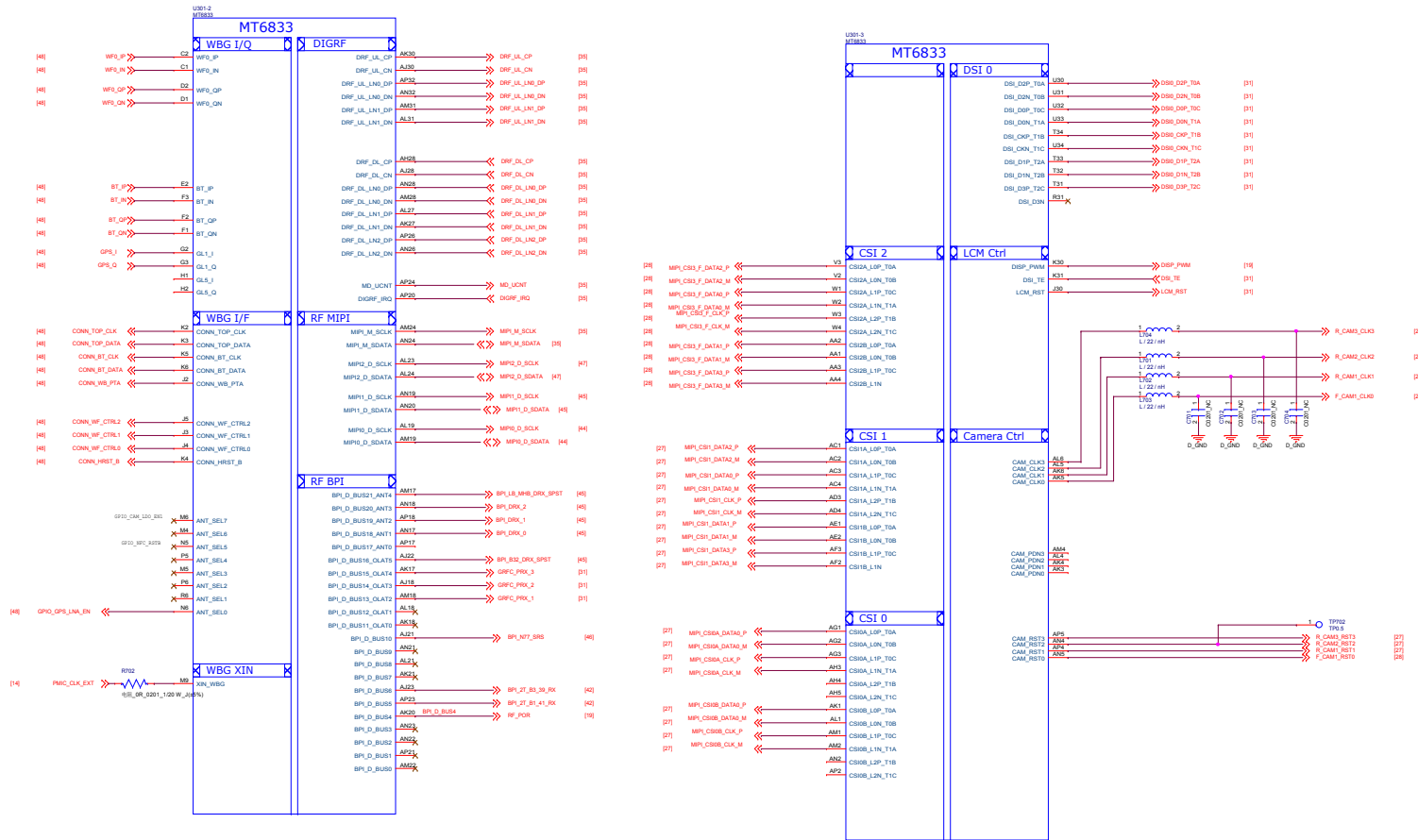
华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title SM6350_Control		
Size D	Project <Doc>	Rev V1
Date: Tuesday, January 12, 2021	Sheet 2 of 48	



华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title 05.BB_POWER_IO		
Size D	Project <Doc>	Rev v1
Date: Tuesday, January 12, 2021		Sheet 5 of 48

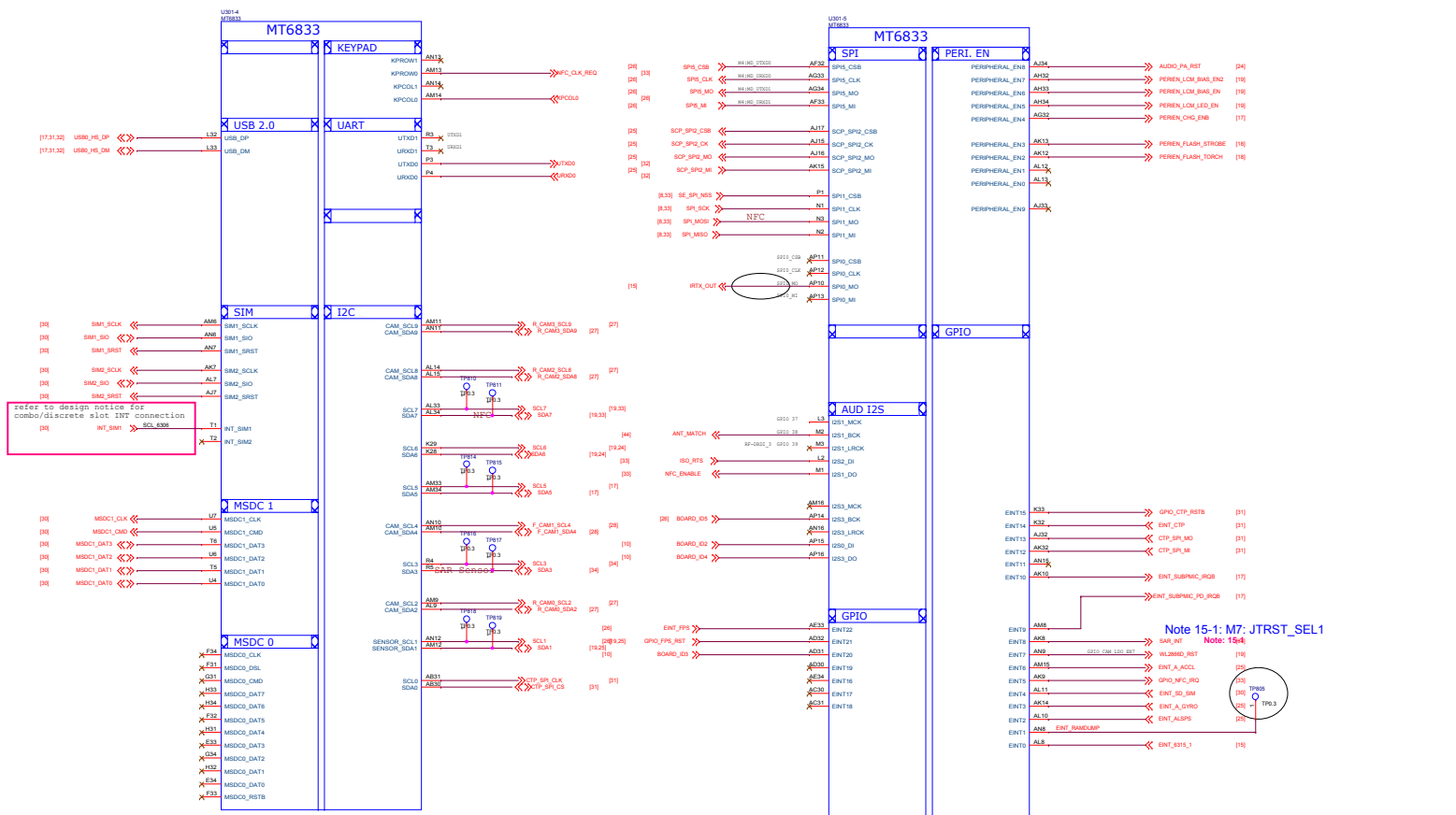


华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title		
06.BB_1		
Size	Project	Rev
D	<Doc>	v1
Date:	Tuesday, January 12, 2021	Sheet 6 of 48

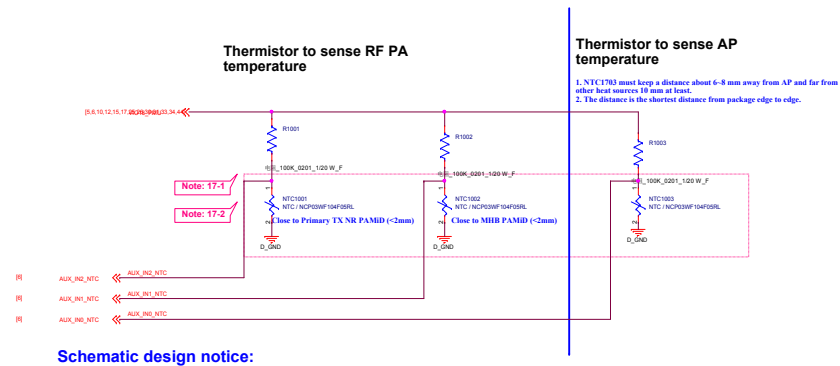


华勤通讯 Huaqin Telecom Technology Com.,Ltd

Title			07.BB_2
Size	Project	<Doc>	
D			Rev v1
Date:	Tuesday, January 12, 2021	Sheet	7 of 48



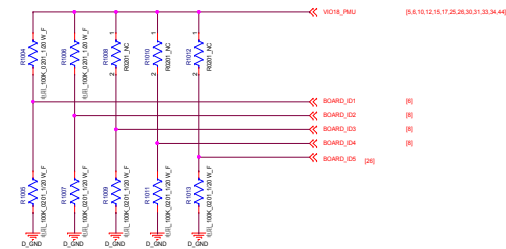
华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title		08.BB_3	
Size	Project	<Doc>	Rev
D			v1
Date:	Tuesday, January 12, 2021	Sheet 8 of 48	



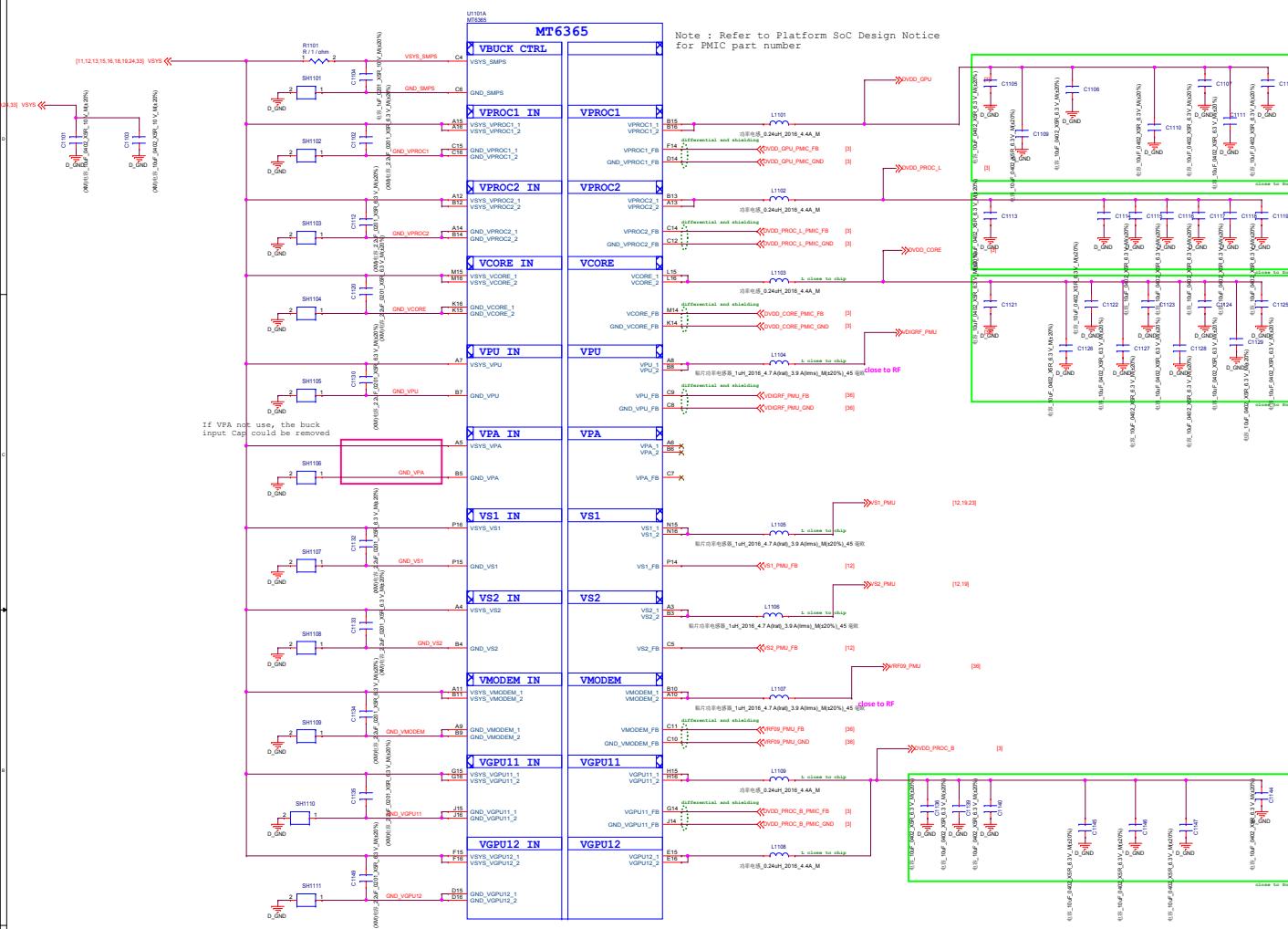
Schematic design notice:

Note 17-1: SW Default Configuration is as followings
AUX_IN0_NTC is for AP Temp.
AUX_IN1_NTC is for MHB PAMID Temp.
AUX_IN2_NTC is for NR PAMID Temp.

Note 17-2: If your design or placement is different with SW Default Config.
Please refer to "MTXXXX Thermal User Manuel.docx" and modify SW setting.



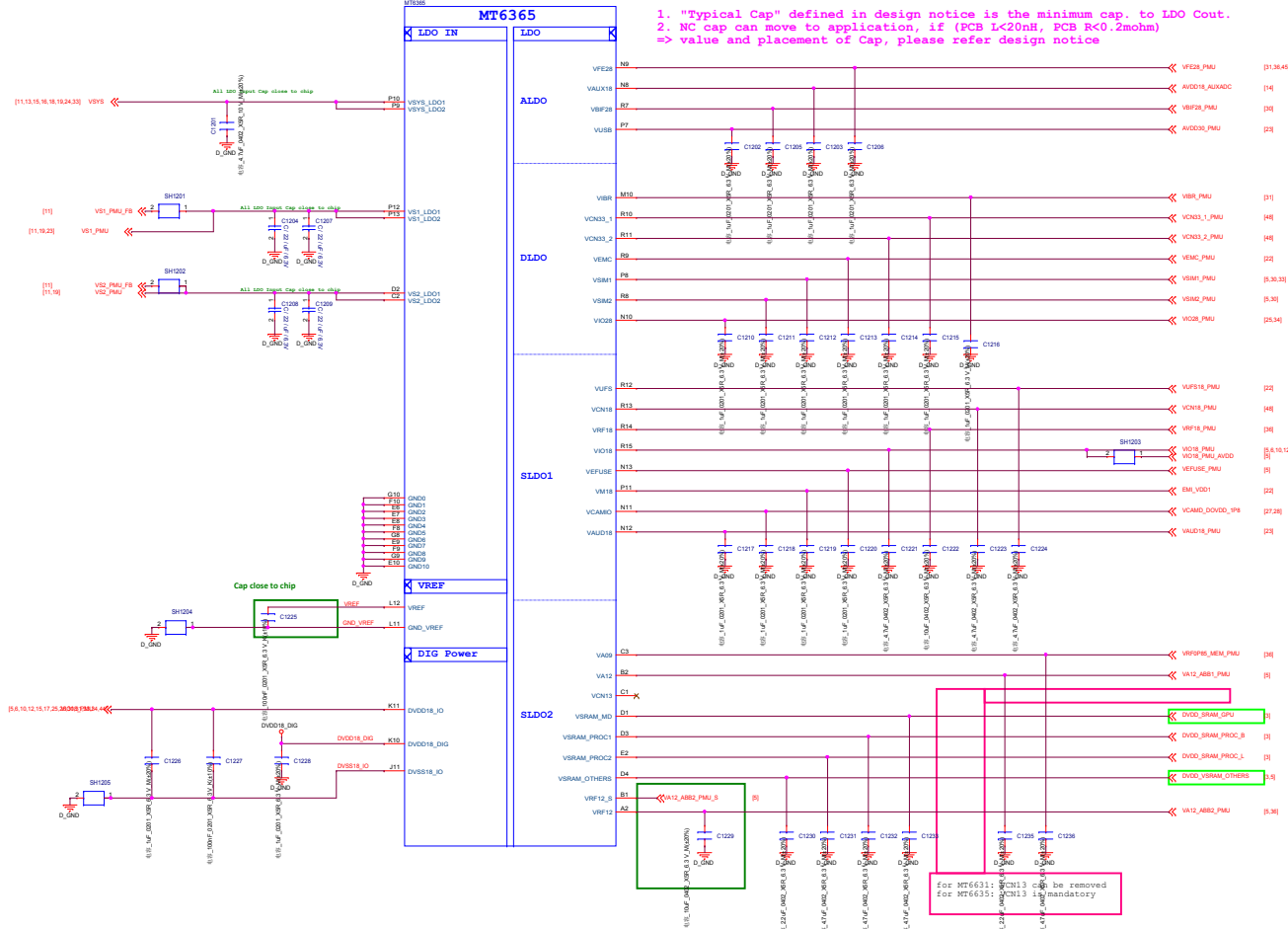
华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title 10.BB_AUXADC_Thermal/BOARD_ID			
Size D	Project <Doc>		Rev v1
Date: Tuesday, January 12, 2021	Sheet 10 of 48		



Circuit Type	Buck Name (Application name)	Output Voltage Range (V)	Boot Default (V)	I _{OUT-MAX} (mA)
Buck	VPROC1 (DVDD_GPU)	0.40 ~ 1.19 (6.25mV/step)	ON (0.75)	4800
	VPROC2 (DVDD_PROC_L)	0.40 ~ 1.19 (6.25mV/step)	ON (0.75)	4800
	VGPU11+12 (DVDD_PROC_B)	0.40 ~ 1.19 (6.25mV/step)	ON (0.75)	4800 *2
	VCORE (DVDD_CORE)	0.40 ~ 1.3 (6.25mV/step)	ON (0.75)	4800
	VMODEM (VRF09_PMU)	0.50 ~ 1.10 (6.25mV/step)	OFF	4800
	VPU (VDIGRF_PMU)	0.40 ~ 1.19 (12.5mV/step)	ON (0.7)	2400
	VS1 (VS1_PMU)	1.86 ~ 2.20 (12.5mV/step)	ON (2.0)	2200
	VS2 (VS2_PMU)	1.20 ~ 1.50 (12.5mV/step)	ON (1.35)	2500
	VPA (VPA_PMU)	0.50 ~ 3.40 (50mV/step)	OFF	1000

华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title		11.POWER_MT6365_Buck	
Size	Project	<Doc>	
D		Rev	
Date: Tuesday, January 19, 2021		Sheet 11 of 48	
		V1	

Note : Refer to Platform SoC Design Notice
for PMIC part number



Circuit Type	LDO Name	Output Voltage (V)	Boot Default (V)	I _{OUT,MAX} (mA)	Expected use
ALDO	VFE28	2.80	OFF (2.8)	200	RF/E
	VUSB	3.07	ON (3.07)	200	USB
	VAUX18	1.84	ON (1.84)	50	AUXADC
	VIO22	2.24	ON (2.24)	25	DCXO
	VBIFF28	2.80	OFF (2.8)	50	Battery Interface
VRTC	VRTC	2.80	ON (2.8)	2	RTC
VDIG	DVDD18_DIG	1.80	ON (1.8)	10	PMIC Digital

Circuit Type	LDO Name	Output Voltage (V)	Boot Default (V)	I _{OUT,MAX} (mA)	Expected use
DLDO	VSIM1	1.7/1.8/1.86/2.76/3.0/3.1	OFF (1.86)	200	SIM
	VSIM2	1.7/1.8/1.86/2.76/3.0/3.1	OFF (1.86)	200	SIM
	VCN33_1	3.3/3.4/3.5/3.6	OFF (3.3)	800	Connectivity
	VCN33_2	3.3/3.4/3.5/3.6	OFF (3.3)	800	Connectivity
	VEMC	2.55/2.9/3.0/3.3	ON (3.0)	800	eMMC / UFS
	VIO28	2.8/2.9/3.0/3.1/3.2/3.3	OFF (2.8)	200	IO & Sensor
	VIBR	2.7/2.8/3.3	OFF (2.8)	200	Vibrator

Circuit Type	LDO Name	Output Voltage (V)	Boot Default (V)	I _{OUT,MAX} (mA)	Expected use
SLD01	VEFUSE	1.80	OFF (1.8)	300	EFUSE
	VAUD18	1.80	ON (1.8)	300	Audio
	VCAMIO	1.80	OFF (1.8)	300	Camera IO
	VM18	1.80	ON (1.8)	300	DRAM
	VRF18	1.80	OFF (1.8)	450	RF
	VIO18	1.80	ON (1.8)	600	IO & Sensor
	VCN18	1.80	OFF (1.8)	1200	Connectivity
	VUFS	1.80	ON (1.86)	1200	eMMC / UFS
	VREF	1.40	ON (1.4)	10	DCXO
	VBBCK	1.24	ON (1.24)	10	DCXO

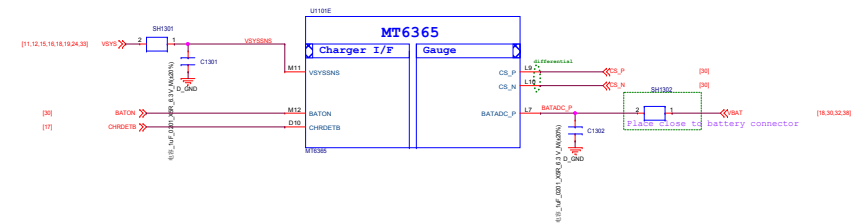
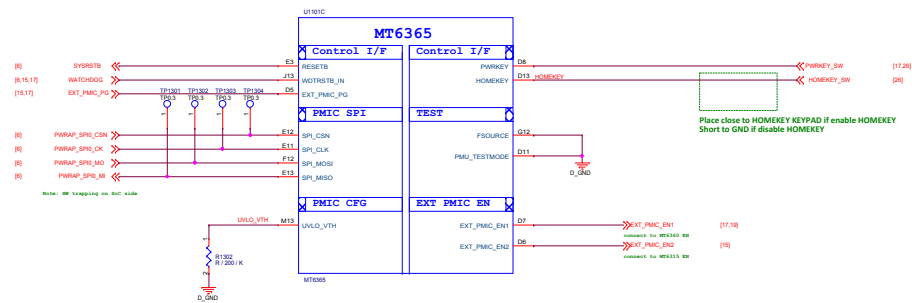
Circuit Type	LDO Name	Output Voltage (V)	Boot Default (V)	I _{OUT,MAX} (mA)	Expected use
SLD02	VA09	0.85	ON (0.85)	300	DIGRF SRAM (VRFOP85_MEM_PMIU)
	VA12	1.2	ON (1.2)	300	AP Analog Module
	VCN13	1.3	OFF (1.3)	350	Connectivity
	VS RAM_PROCI	0.60 ~ 1.2 (6.25mV/step)	ON (0.85)	600	CPUB SRAM
	VS RAM_PROCI2	0.60 ~ 1.2 (6.25mV/step)	ON (0.85)	600	CPUL SRAM
	VS RAM_PROCI3	0.60 ~ 1.2 (6.25mV/step)	ON (0.85)	600	OTHERS SRAM
	VS RAM_PROCI4	0.60 ~ 1.2 (6.25mV/step)	ON (0.85)	600	GPU SRAM
	VS RAM_PROCI5	0.60 ~ 1.2 (6.25mV/step)	ON (0.85)	600	GPU SRAM
	VRF12	1.2	ON (1.2)	800	AP Analog Module

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Title 12.POWER_MT6365_LDO

Size D Project <Doc> Rev V1

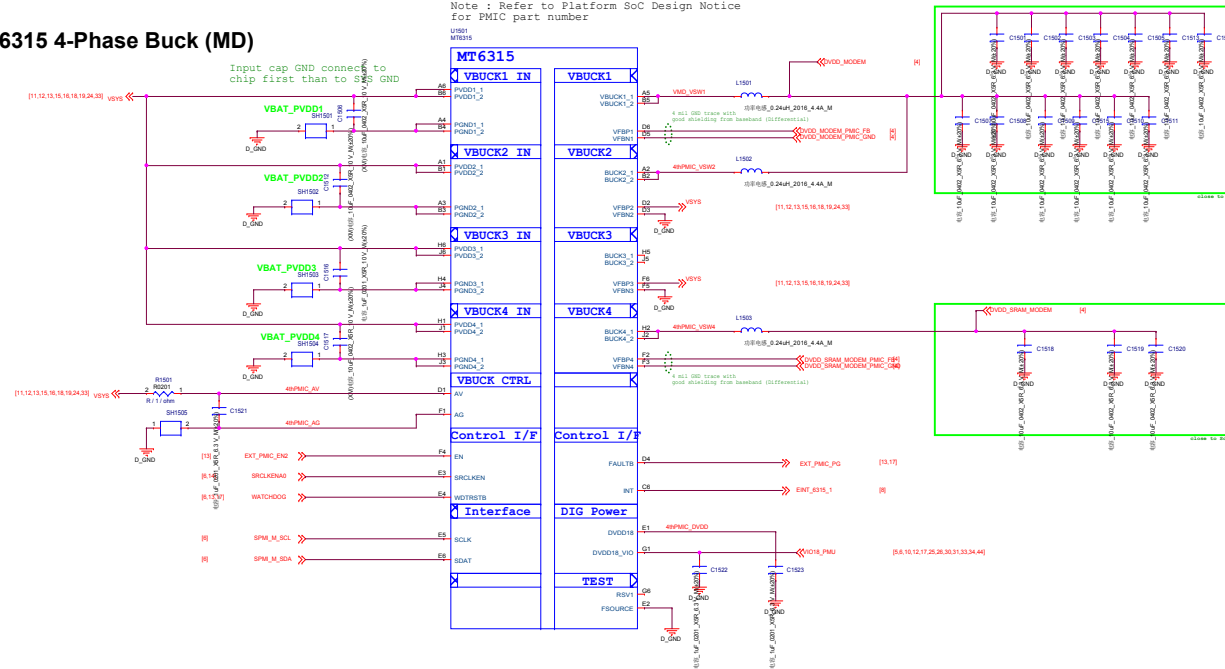
Date: Tuesday, January 12, 2021 Sheet 12 of 48



华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title		13.POWER_MT6365_General	
Size	Project	<Doc>	Rev
D			V1
Date:	Tuesday, January 12, 2021	Sheet	13 of 48

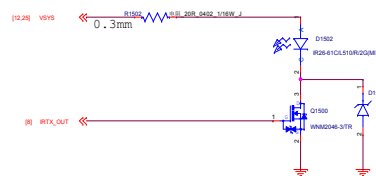
MT6315 4-Phase Buck (MD)

Note : Refer to Platform SoC Design Notice for PMIC part number

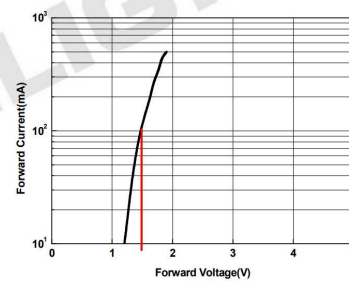


MT6315NP/B	Output Voltage Range (V)	Boot Default (V)	I _{OUT-MAX} (mA)	Expected use
BUCK1+2	0.40 ~ 1.19375 (6.25mV/step)	ON (0.825)	5000*2	DVDD_MODEM
BUCK3	0.40 ~ 1.19375 (6.25mV/step)	ON (0.75)	5000	—
BUCK4	0.40 ~ 1.19375 (6.25mV/step)	ON (0.825)	5000	DVDD_SRAM_MODEM

IR LED



Forward Current vs. Forward Voltage



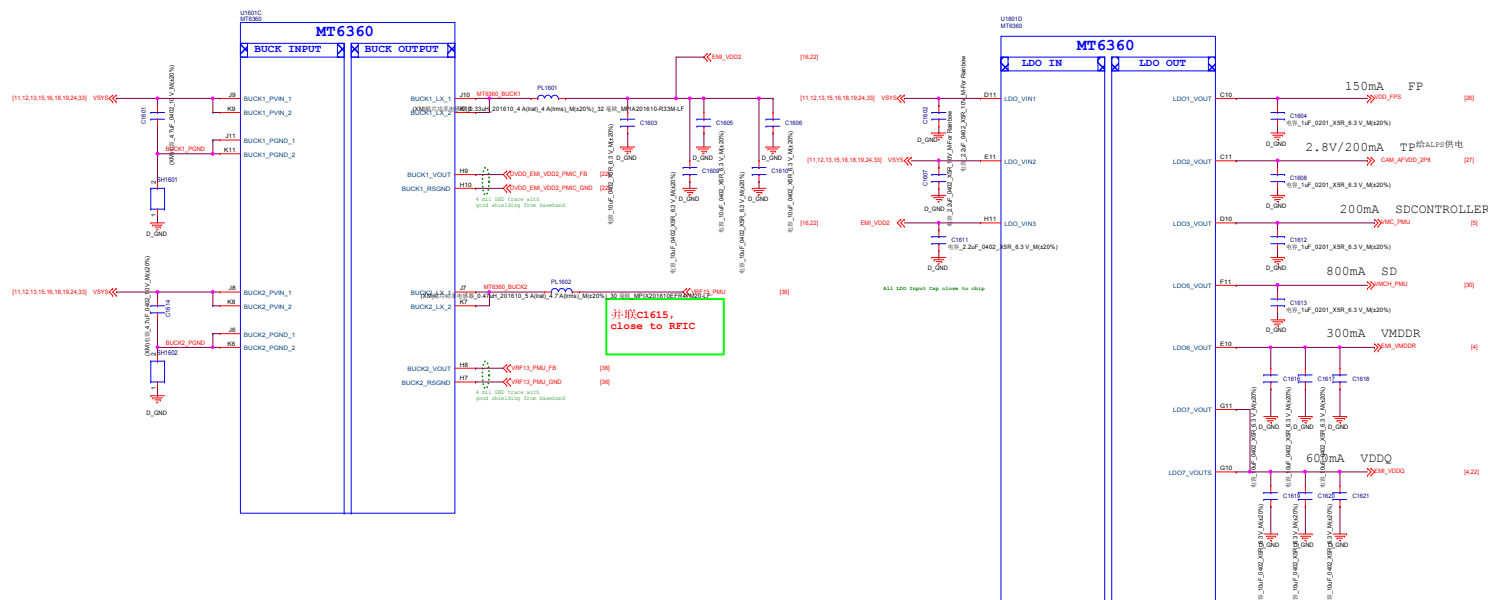
华勤通讯 Huaqin Telecom Technology Com.,Ltd

Title 15.POWER_MT6315_VMD/IR_LED

Size D Project <Doc> Rev V1

Date: Tuesday, January 12, 2021 Sheet 15 of 48

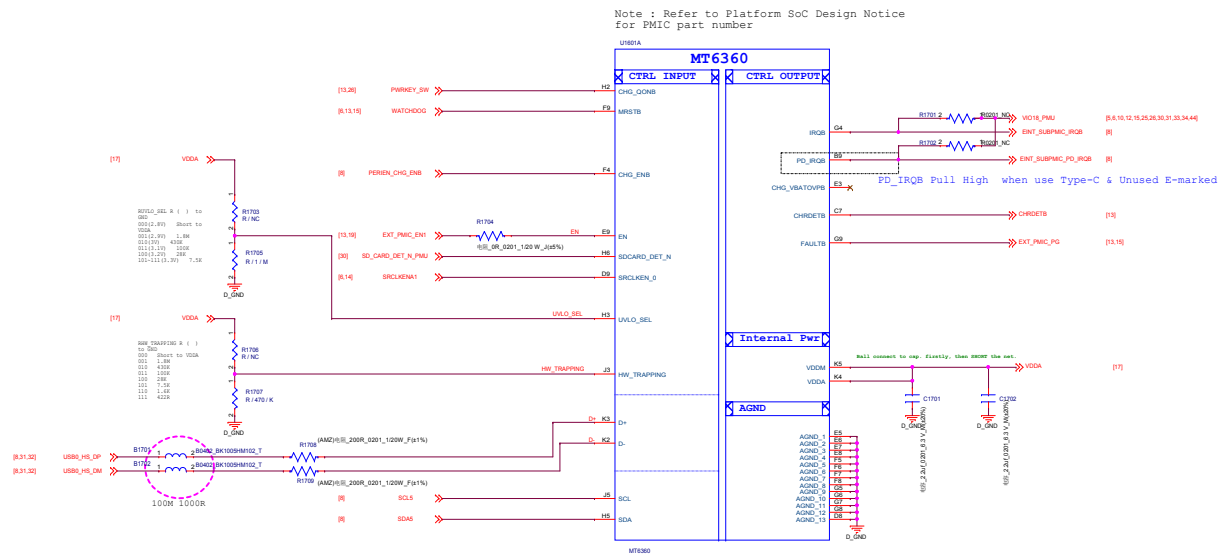
Note : Refer to Platform SoC Design Notice
for PMIC part number



Circuit Type	Name	Output Voltage Range (V)	Boot Default (V)	I _{OUT-MAX} (mA)	Expected use
Buck	BUCK1	0.3 ~ 1.3 (5mV/step)	ON (1.125)	3000	EMI_VDD2
	BUCK2	0.3 ~ 1.3 (5mV/step)	OFF (1.3)	3000	VRF13_PMU

Circuit Type	Name	Output Voltage (V)	Boot Default (V)	I _{OUT-MAX} (mA)	Expected use
LDO	LDO1	1.8/2.0/2.1/2.5/2.7/ 2.8/2.9/3.0/3.1/3.3	OFF (1.8)	150	Fingerprint
	LDO2	1.8/2.0/2.1/2.5/2.7/ 2.8/2.9/3.0/3.1/3.3	OFF (1.8)	200	Touch Panel
	LDO3	1.8/2.9/2.8/3.0/3.3	OFF (3.0)	200	SD Card
	LDO5	2.9/3.0/3.3	OFF (3.0)	800	SD Card
	LDO6	0.75	ON (0.75)	300	EMI_VMDDR
	LDO7	0.6	ON (0.6)	600	EMI_VDDQ

华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title 16.POWER_MT6360_BUCK_LDO*		
Size D	Project <Doc>	Rev v1
Date: Tuesday, January 12, 2021	Sheet 16 of 48	



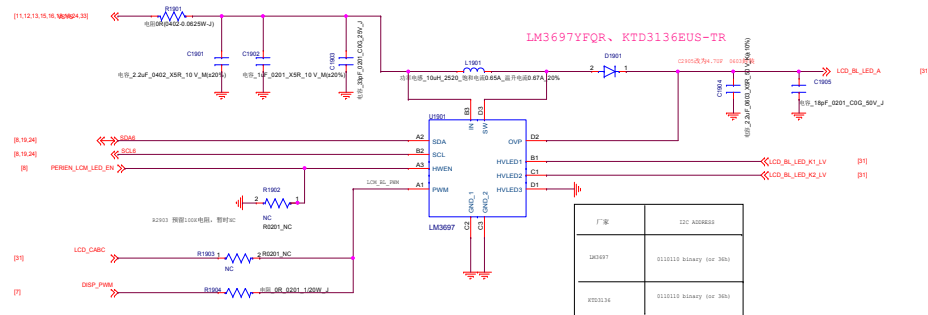
华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title 17.POWER_MT6360_General		
Size D	Project <Doc>	Rev v1
Date: Tuesday, January 12, 2021	Sheet 17 of 48	

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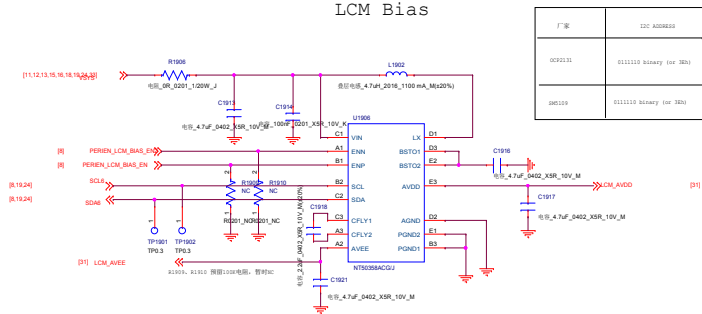
Note 28-1: For better ESD & surge performance we need choose suitable device for system protection. Please refer to the latest version of [Surge device selection guide] provided by MTK.

华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title 18.POWER_MT6360_Charger			
Size D	Project <Doc>		Rev V1
Date: Tuesday, January 12, 2021		Sheet 18 of 48	

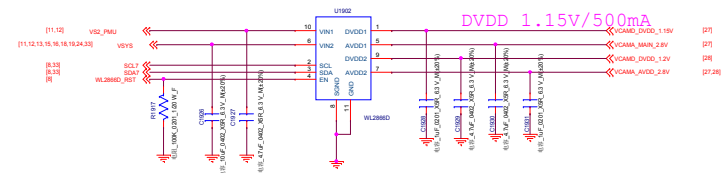
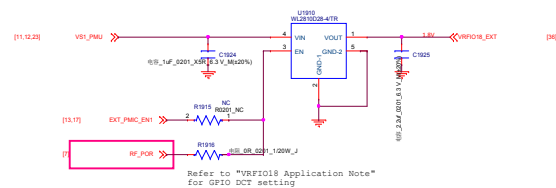
LCM Backlight LED Driver



LCM Bias



External VIO18_LDO: option for power leakage from UART port on product line



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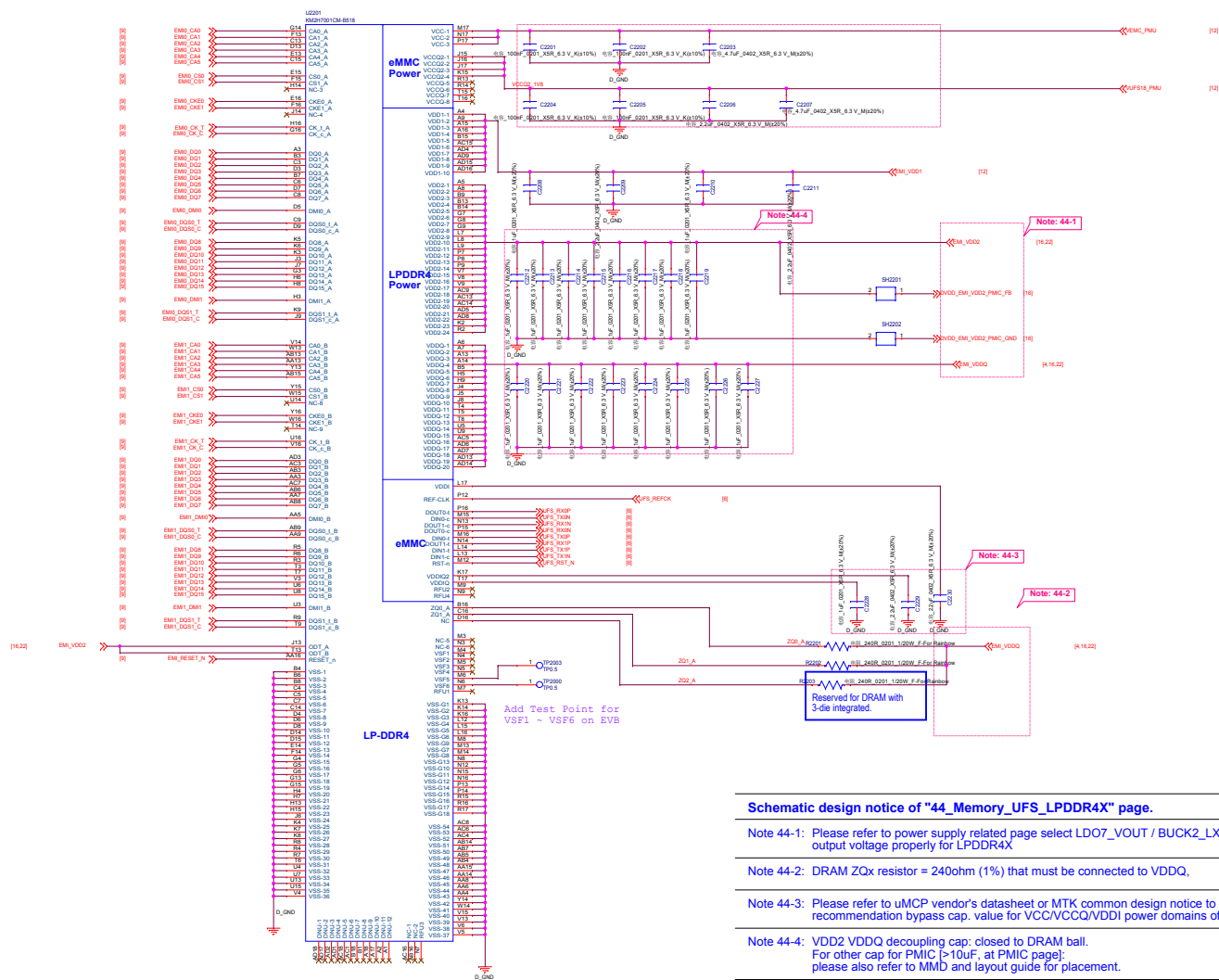
Title 19.POWER_EXT

Size	Project	Rev
D	<Doc>	V1

Date: Tuesday, January 12, 2021 Sheet 19 of 48

华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title PM7250B_CHARGE/FG		
Size E	Project <Doc>	Rev v1
Date: Tuesday, January 12, 2021	Sheet 20 of 48	

华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title PM7250B_MISC/GPIO/GND		
Size D	Project <Doc>	Rev v1
Date: Tuesday, January 12, 2021		Sheet 21 of 48



Schematic design notice of "44_Memory_UFS_LPDDR4X" page.

Note 44-1: Please refer to power supply related page select LDO7_VOUT / BUCK2_LX output voltage properly for LPDDR4X

Note 44-2: DRAM ZQx resistor = 240ohm (1%) that must be connected to VDDQ.

Note 44-3: Please refer to uMCP vendor's datasheet or MTK common design notice to get the recommendation bypass cap. value for VCC/VCCQ/VDD1 power domains of UFS.

Note 44-4: VDD2 VDDQ decoupling cap: closed to DRAM ball.
For other cap for PMIC (>10uF, at PMIC page); please also refer to MMD and layout guide for placement.

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Title
22.Memory_UFS_LPDDR4X

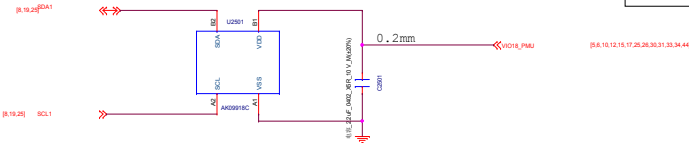
Size D	Project <Doc>	Rev v1
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Date: Tuesday, January 12, 2021 Sheet 22 of 48

eCOMPASS

M-Sensor

厂家	I2C ADDRESS
AK8963C	0ch
AK8963C	2ch

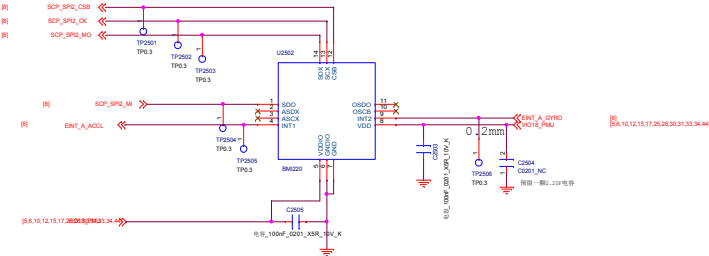


A+Gsensor

default SPI (if sensor hub support)

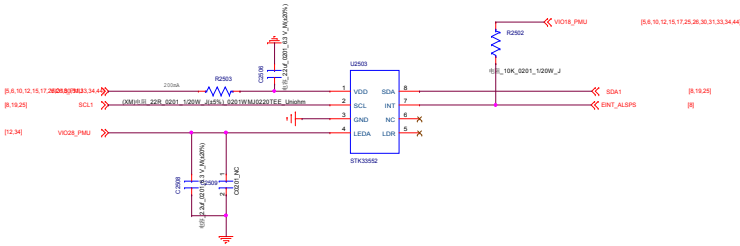
厂家	I2C ADDRESS
AK8963C	
AK8963C	

Gyro+A sensor



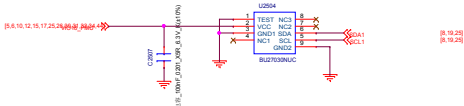
ALS+PS

厂家	I2C ADDRESS
STK3352	0ch



后光感 BU27030NUC

厂家	I2C ADDRESS
BU27030NUC	0ch

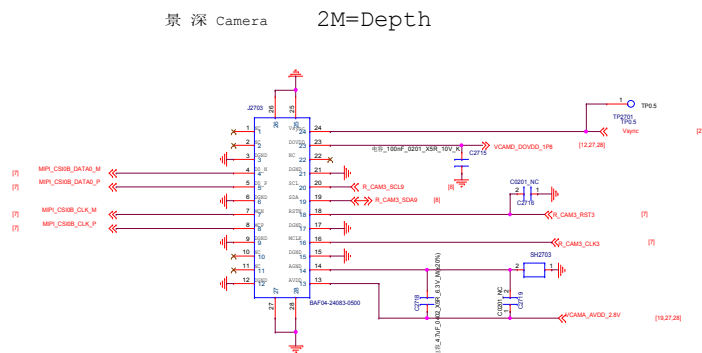
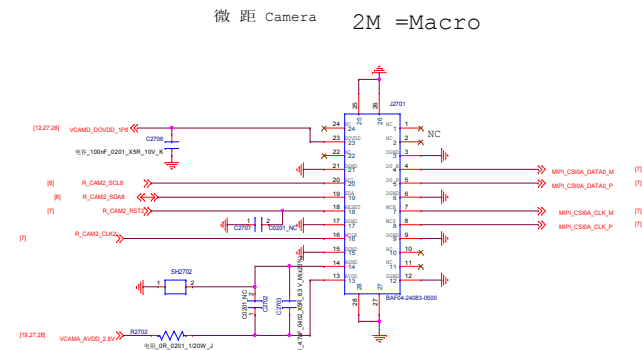
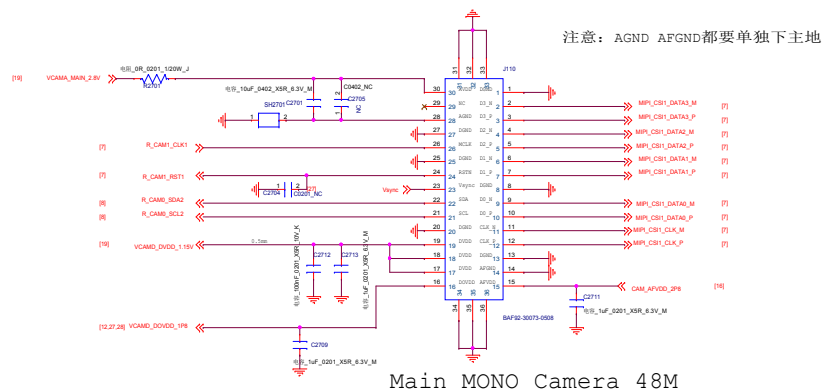


华勤通讯 Huaqin Telecom Technology Com.,Ltd

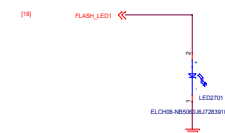
Title 25.SENSOR

Size D Project <Doc> Rev v1

Date: Tuesday, January 12, 2021 Sheet 25 of 48



Flash LED 5V Boost



Flash 下方地要完整且足够大, 散热一定要好

华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title		27.REAR_CAM	
Size	Project	<Doc>	Rev
D			V1
Date: Tuesday, January 12, 2021		Sheet 27 of 48	

华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title		Sub_Charger_SMB1396	
Size D	Project <Doc>		Rev v1
Date: Tuesday, January 12, 2021		Sheet 29 of	48

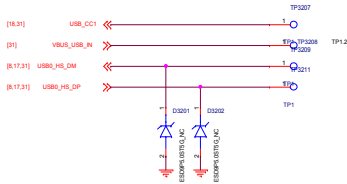
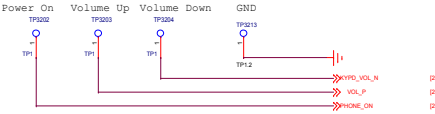
VBAT TEST POINT



VBAT

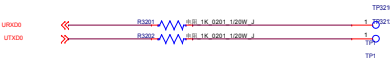
KEYPAD

KEY BTB

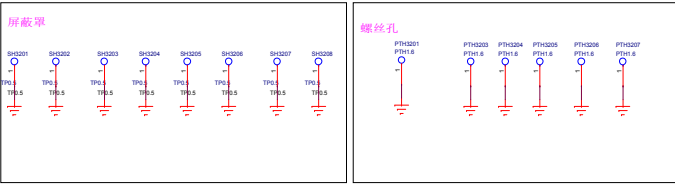


VCHG

UART TP

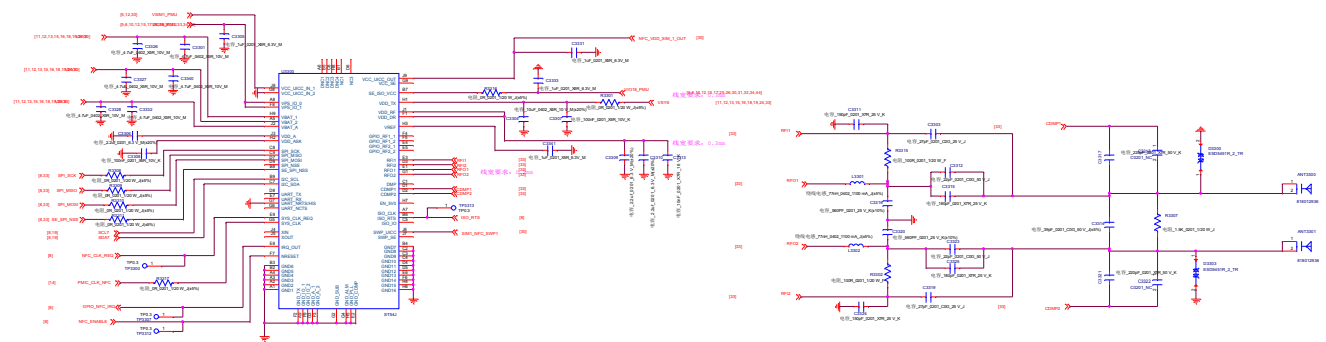


JTAG

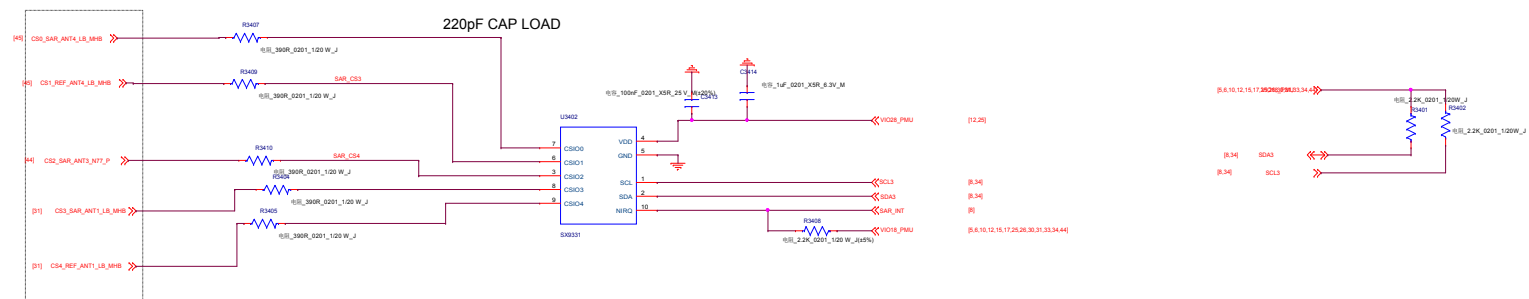


华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title 32.TEST_POINT/SHIELD		
Size D	Project <Doc>	Rev v1
Date: Tuesday, January 12, 2021		Sheet 32 of 50

RF_NFC_not support



SAR_sensor Not support



Design notice:

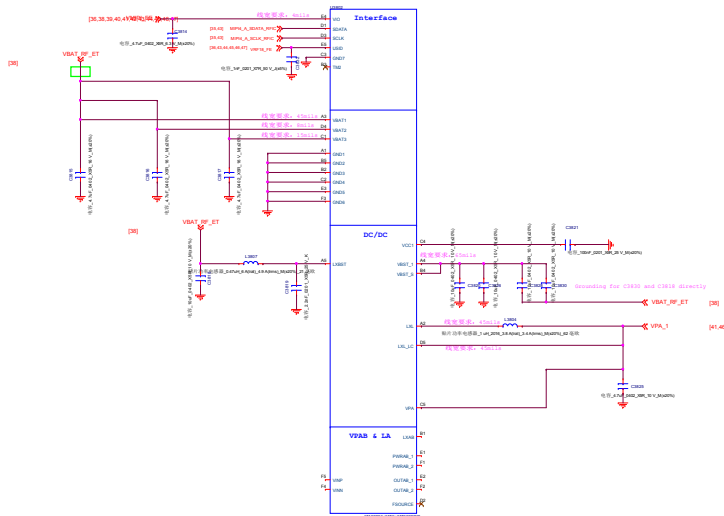
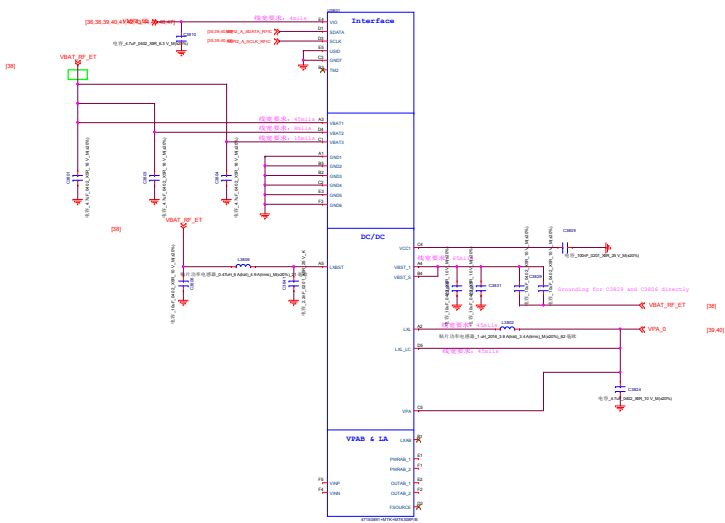
- 1) CS0_1 nets are differential pairs;
- 2) The width for each line, should be less than 0.1mm;
- 3) 0.2mm should be guaranteed between the ground and differential pairs;
- 4) And also, the distance between the differential line should be more than 0.2mm;
- 5) All the CSx nets should be protected by surrounding GND;

华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title SAR_sensor			
Size D	Project <Doc>		Rev v1
Date:	Tuesday, January 12, 2021	Sheet	34 of 48

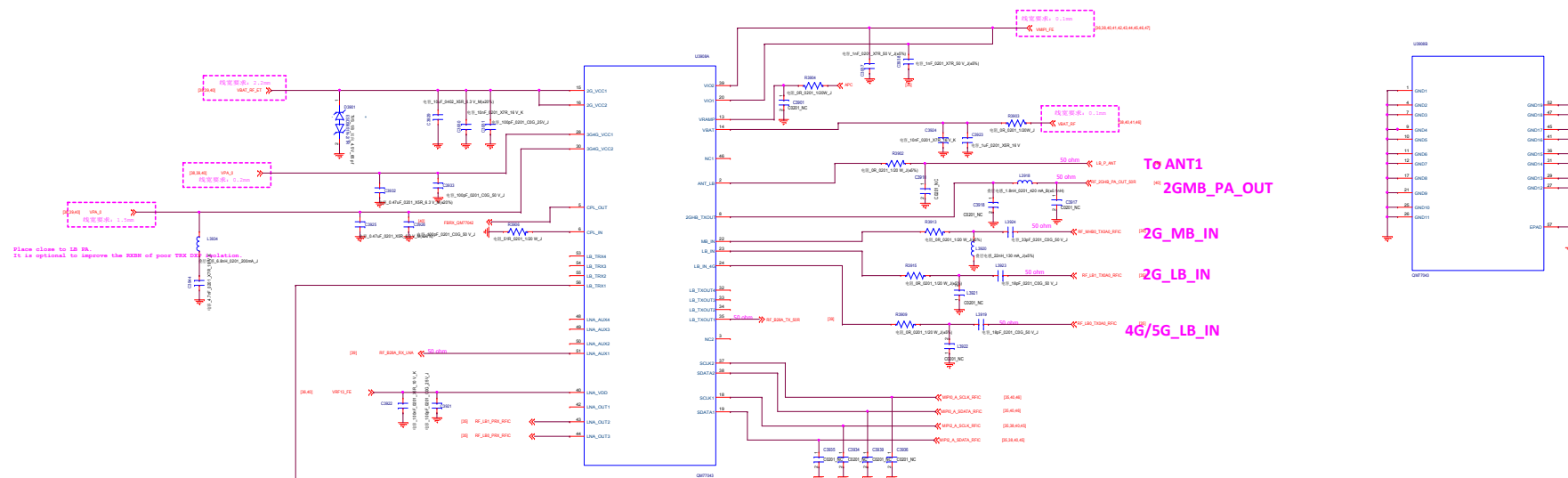
华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title		NC_FBRX	
Size D	Project <Doc>		Rev v1
Date: Tuesday, January 12, 2021		Sheet 37 of	48

QM77042 and QM77048 for one APT

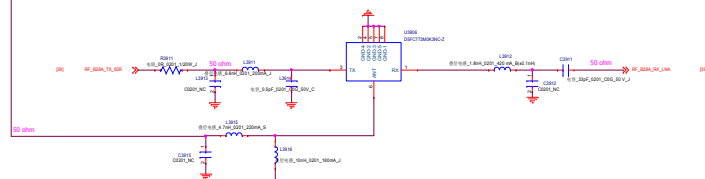
VC7643 and QM78200 for the other APT



2G_3G/4G/5G LB PA_QM77042



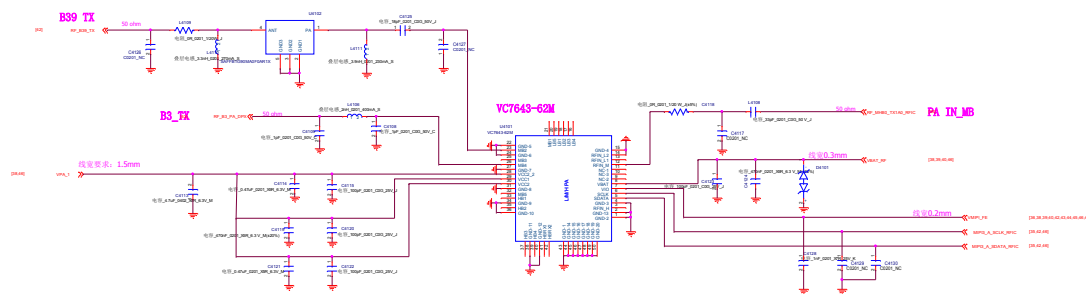
QM77042同QM77043 pin to pin



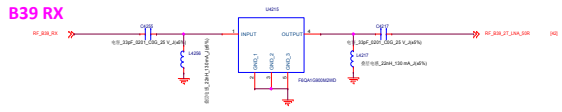
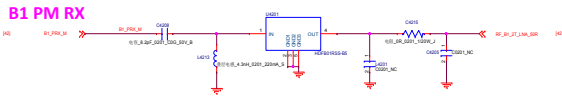
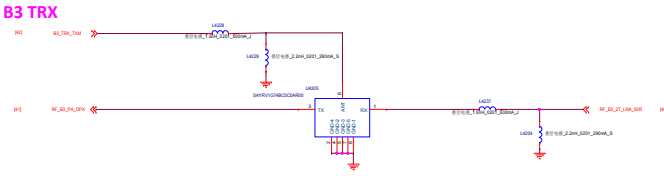
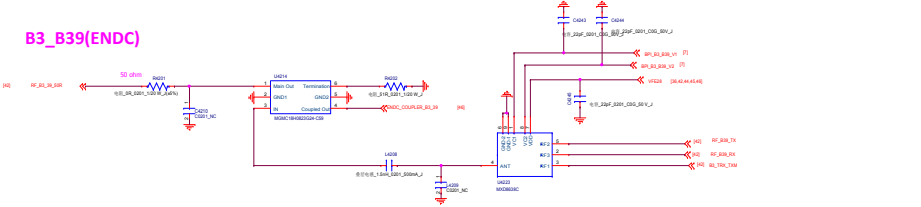
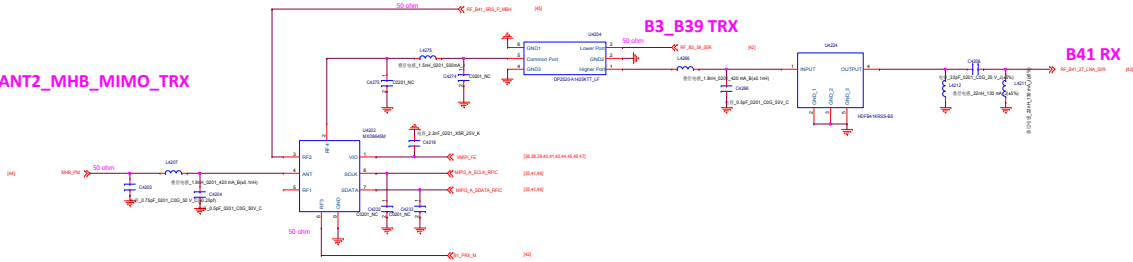
NTC温度检测
见P10的R1001 and NTC1001

华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title QM77042			
Size AQ	Project <Doc>	Rev v1	
Date: Tuesday, January 12, 2021	Sheet 39 of 48		

MMMB PA_MB ENDC

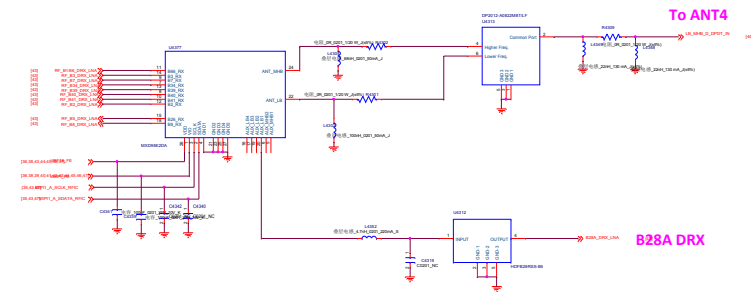
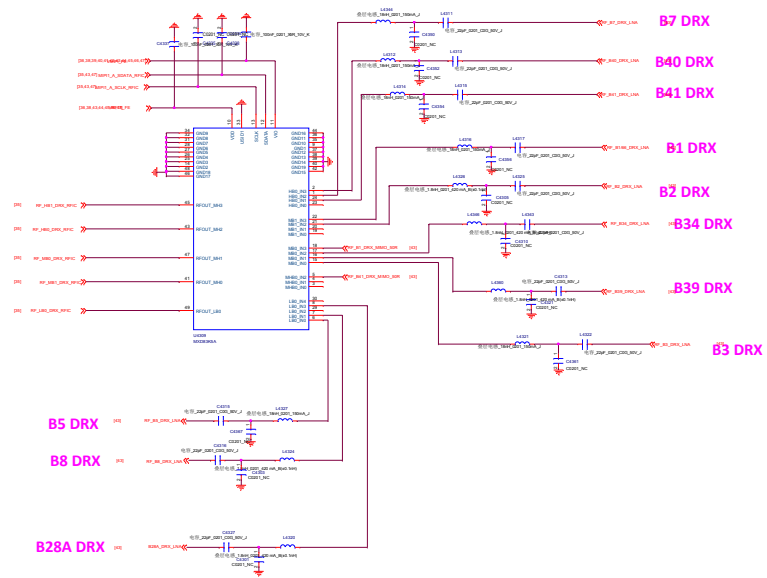


MMMB PA_VC7643_ENDC_MHB_TRX

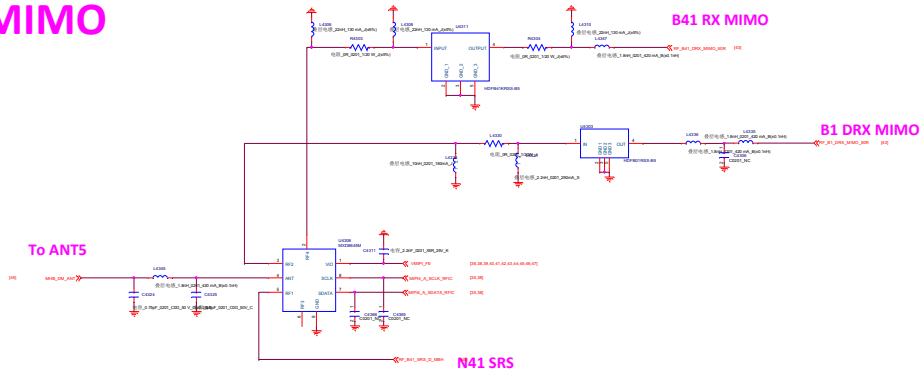


华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title MMMB PA_VC7643_ENDC_MHB_TRX			
Size A0	Project <Doc>		Rev v1
Date: Tuesday, January 12, 2021		Sheet 42 of 48	

DRX_MXD98E2DA+MXD83K5A

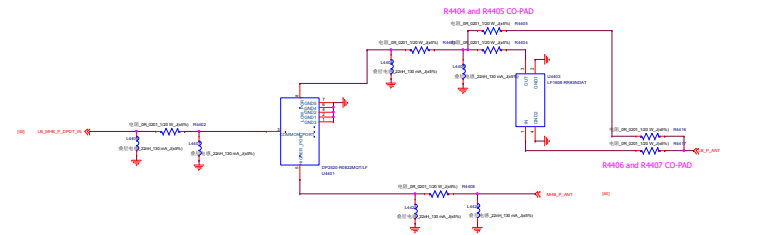
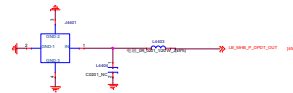
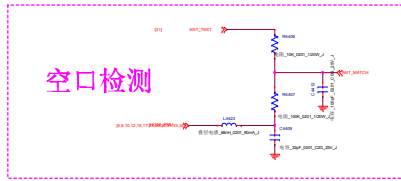


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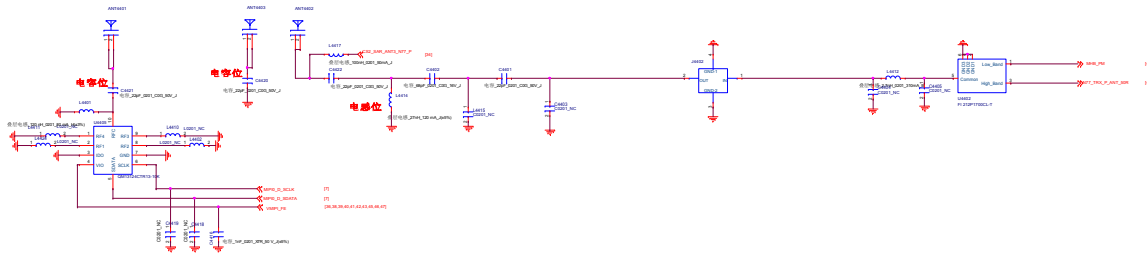


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Title			
MXD98E2DA_MXD83K5A			
Size	Project	<Doc>	Rev
E			V1
Date: Tuesday, January 12, 2021		Sheet 43 of 48	

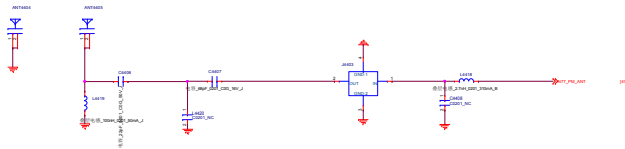
ANT1 (LB_MHB_P) to 小板



ANT2 (MHB_PM/N77_TRX)

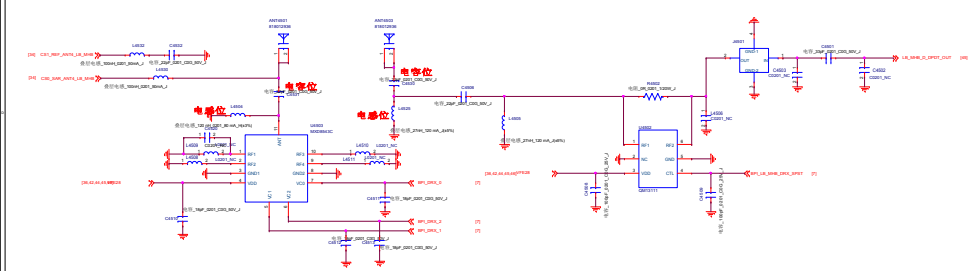


ANT3 (N77_PM)

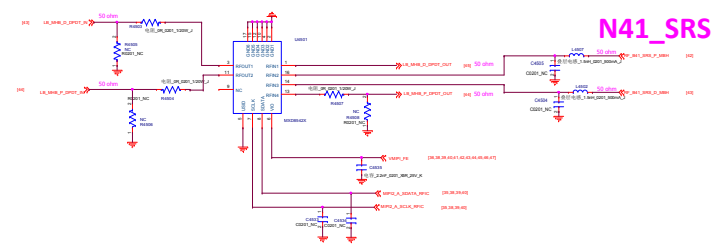


华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title			
ANT_P_PM			
Size	Project	<Doc>	Rev
E			V1
Date: Tuesday, January 12, 2021		Sheet 44 of 48	

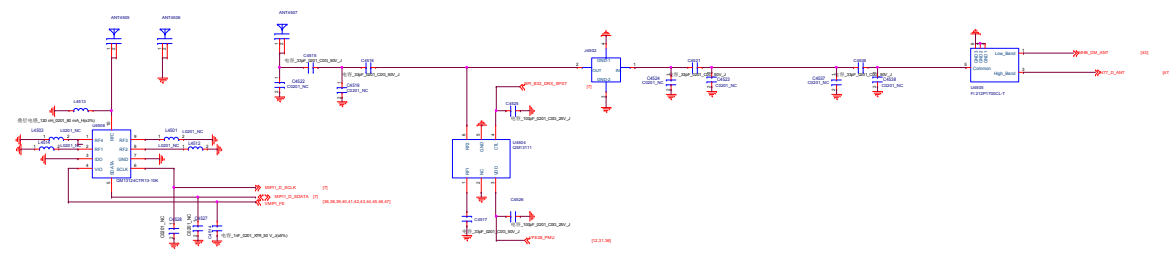
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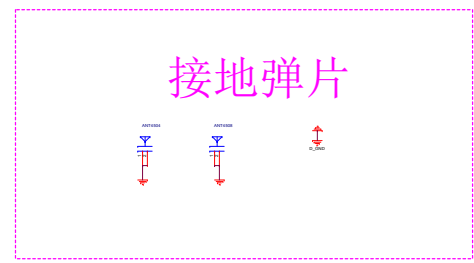
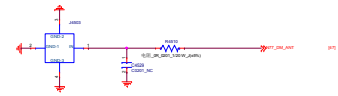
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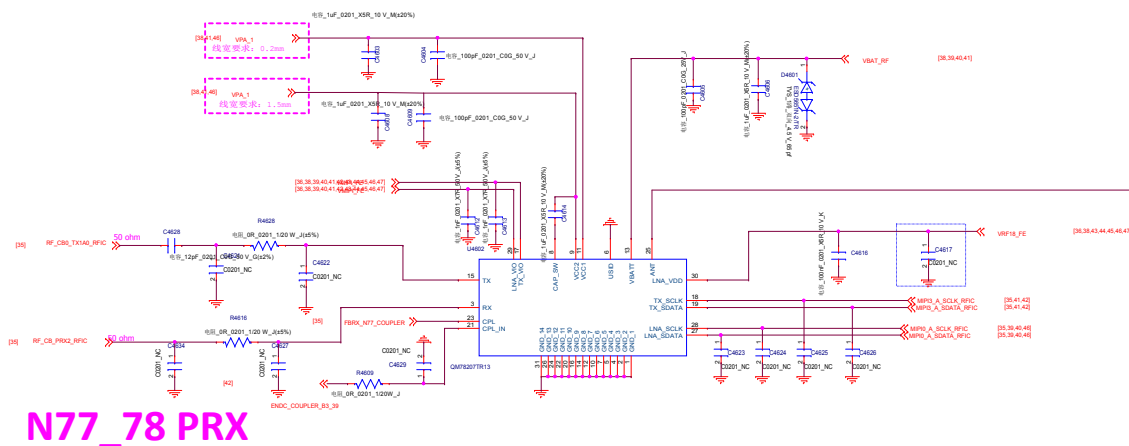


ANT6 (N77_DM) to 小板



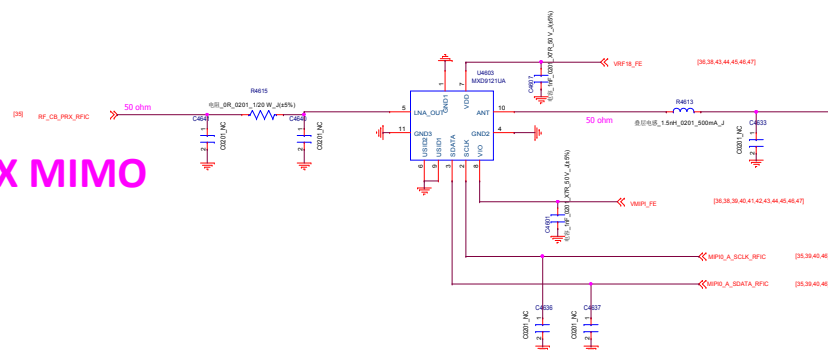
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Title ANT_D_DM			
Size E	Project	<Doc>	Rev V1
Date: Tuesday, January 12, 2021		Sheet 45 of	48

N77/78_TRX0_QM78207



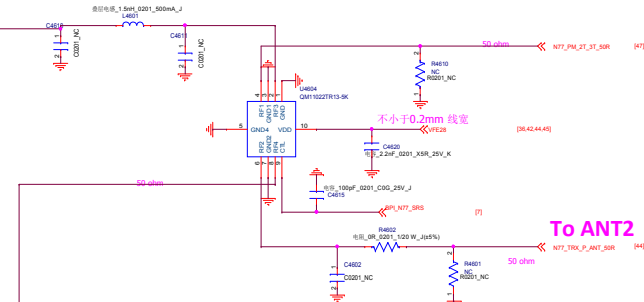
N77/78_MIMO_MXD9121UA

N77_78 PRX MIMO



NTC温度检测
见P10的R1002 and NTC1002

N77/78_SRS



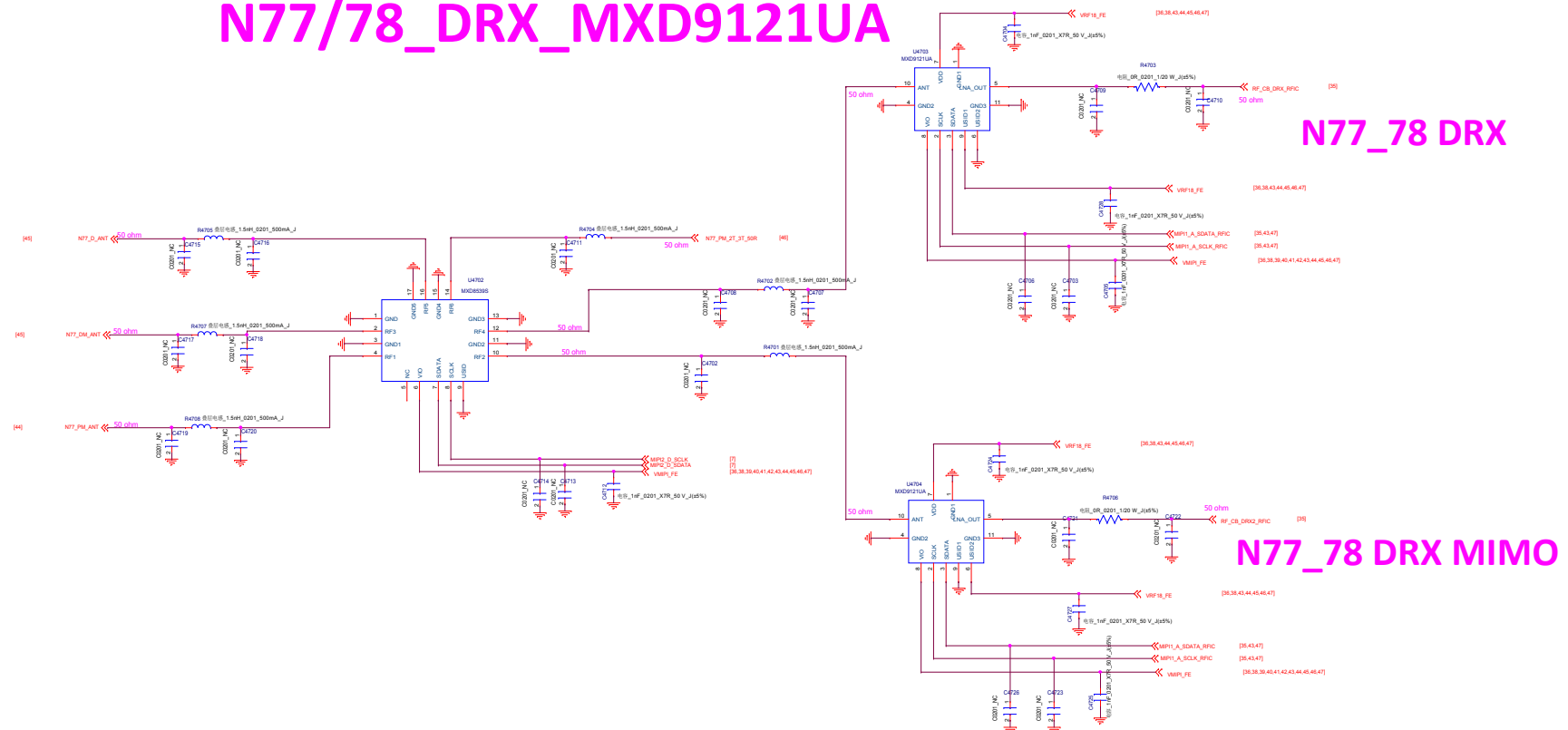
华勤通讯 Huaqin Telecom Technology Com.,Ltd

Title
N77/78_TRX0_QM78207

Size	Project	Rev
D	<Doc>	V1

Date: Tuesday, January 12, 2021 Sheet 46 of 48

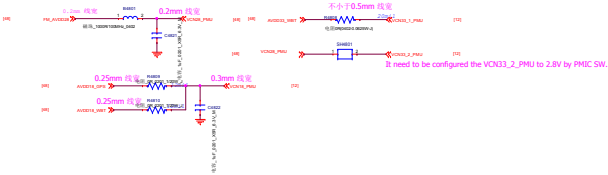
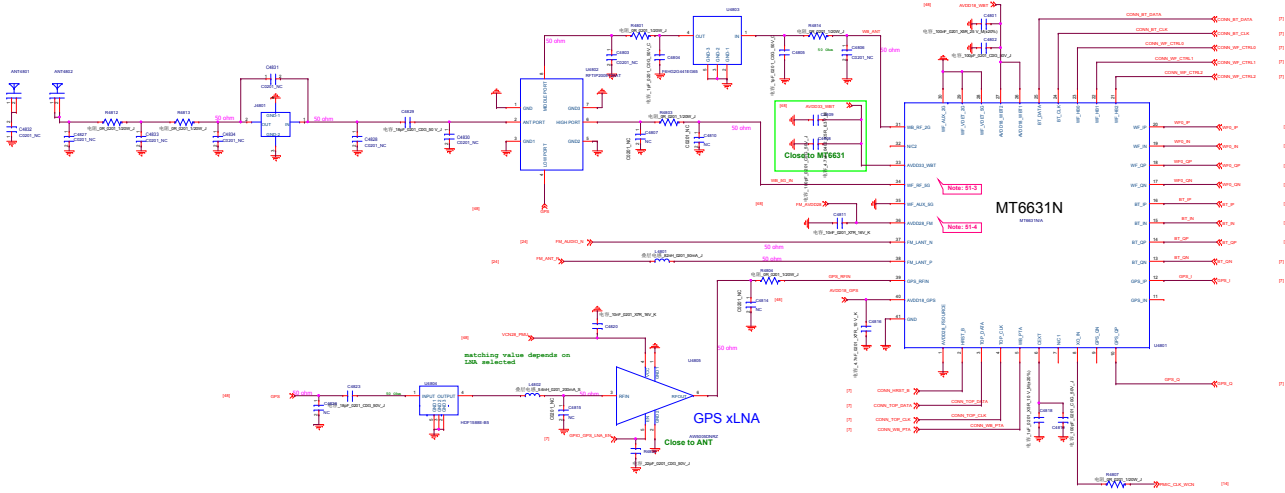
N77/78_DRX_MXD9121UA



华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title N77/78_DRX_MXD9121UA			
Size D	Project <Doc>		Rev v1
Date: Tuesday, January 12, 2021	Sheet 47 of 48		

ANT7
WiFi 2.4G & 5G_GPS

MT6631



Schematic design notice of "S1_CONNECTIVITY_CONSYS_MT6631"

Note 51-1: For RS015 size, please select 0402 size or larger one

Note 51-2: Please refer to MT6765 baseband design notice for VCN33_LDO selection guide

Note 51-3: If WiFi 5G not support, connect pin 34(WF_RF_5G) to GND

Note 51-4: Pin 36 (AVDD28_FM) must be connected to VCN28 even if FM not support